

CSS Grid Layout

Introduction Guide

Master the most powerful CSS layout system

Easy English • 40+ Examples • Ready to Use

Contents

1	Introduction to CSS Grid	2
1.1	What is CSS Grid?	2
1.2	Why Use CSS Grid?	2
1.3	Grid vs Flexbox	3
1.4	Browser Support	3
2	Grid Terminology	4
2.1	Grid Container	4
2.2	Grid Items	4
2.3	Grid Lines	4
2.4	Grid Tracks	4
2.5	Grid Cells	4
2.6	Grid Areas	5
3	Basic Grid Setup	6
3.1	Creating a Simple Grid	6
3.2	Flexible Units	6
4	Summary	7

Chapter 1

Introduction to CSS Grid

1.1 What is CSS Grid?

CSS Grid Layout is a two-dimensional layout system that allows you to create complex layouts using rows and columns. It's a revolutionary tool for web design that gives you complete control over positioning elements on a page without relying on floats, flexbox, or positioning hacks.

Think of it as: "A powerful 2D layout system with rows and columns, perfect for complex page layouts."

[colback=lightblue, colframe=codeblue, coltitle=darkblue, title=Key Concept] CSS Grid lets you define a grid structure with rows and columns, then place items anywhere on that grid with precision and flexibility.

1.2 Why Use CSS Grid?

- Perfect for 2D layouts (rows + columns)
- Precise item positioning
- Responsive design without media queries
- Less code than float-based layouts
- Better browser support than older methods
- Works with content reordering
- Easy alignment options
- Ideal for complex page layouts

1.3 Grid vs Flexbox

Feature	Grid	Flexbox
Dimensions	2D (rows + columns)	1D (row or column)
Use Case	Page layouts	Component layouts
Alignment	Both axes	Main + cross axes
Complexity	Advanced layouts	Simple layouts

1.4 Browser Support

CSS Grid has excellent support in all modern browsers:

- Chrome 57+
- Firefox 52+
- Safari 10.1+
- Edge 16+
- Opera 44+

Chapter 2

Grid Terminology

2.1 Grid Container

The parent element with `display: grid`:

[colback=codegray, colframe=darkblue, title=HTML]

```
<div class="grid-container">
  <div class="item">Item 1</div>
  <div class="item">Item 2</div>
</div>
```

[colback=codegray, colframe=darkblue, title=CSS]

```
.grid-container {
  display: grid;
  /* This is the grid container */
}
```

2.2 Grid Items

Direct children of the grid container.

2.3 Grid Lines

Dividing lines creating the structure. Numbered from 1 upward.

2.4 Grid Tracks

Rows and columns created by grid lines.

2.5 Grid Cells

Individual spaces between grid lines.

2.6 Grid Areas

Rectangular groups of cells.

Chapter 3

Basic Grid Setup

3.1 Creating a Simple Grid

[colback=codegray, colframe=darkblue, title=CSS]

```
.container {  
    display: grid;  
    grid-template-columns: 200px 200px 200px;  
    grid-template-rows: 100px 100px;  
    gap: 10px;  
}
```

This creates:

- 3 columns of 200px each
- 2 rows of 100px each
- 10px gap between items

3.2 Flexible Units

[colback=codegray, colframe=darkblue, title=CSS]

```
/* Using fr (fraction) units */  
.container {  
    display: grid;  
    grid-template-columns: 1fr 1fr 1fr;  
    /* Three equal columns */  
}  
  
/* Mix fixed and flexible */  
.container {  
    grid-template-columns: 200px 1fr 200px;  
    /* 200px, flexible middle, 200px */  
}
```

Chapter 4

Summary

CSS Grid is a powerful 2D layout system perfect for creating complex page layouts. Understanding the basics—grid container, grid items, tracks, lines, and cells—is essential before diving into advanced features.

Start practicing by creating simple grids and gradually increase complexity!